MODULES OVER A TOPOLOGICAL GROUP FORM A CATEGORY WITH FUNCTORIAL INJECTIVE EMBEDDINGS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument; the functor is displayed with its definition and the hypothesis needed for the transfer result is verified in the span.

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Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- Modules over a topological group. Etale Cohomology, Definition of an R-G-module and its open stabilizer condition. Supplies the category whose functorial injective embeddings are constructed.
- The functor of invariants and its adjoint. Same chapter, the construction of the functors relating the category to modules over the ring. Supplies the pair of functors used in the proof and the adjunction between them.
- Transfer of functorial injective embeddings. Homology, Lemma homology-lemma-adjoint-functorial-injectives. Supplies that an adjoint pair with the stated hypotheses transfers functorial injective embeddings, which is how the conclusion is reached.

Source reference. Original definition 04JP: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 8786–8786. Navigation pointer invoked at proof line 630; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 0161: sources/stacks-homology/pinned-original-source.tex, source lines 7581–7581. Navigation pointer invoked at proof line 646; the counted span contains the pointer only and the cited result is not reproduced

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Lemma 0.1. *Let G be a topological group. Let R be a ring. The category $Mod_{R,G}$ of R-G-modules, see Étale Cohomology, Definition 04JP, has functorial injective embeddings. In particular this holds for the category of discrete G-modules.*

Proof. By the remark above the lemma the category $\mathrm{Mod}_{R[G]}$ has functorial injective embeddings. Consider the forgetful functor $v : \mathrm{Mod}_{R,G} \to \mathrm{Mod}_{R[G]}$. This functor is fully faithful, transforms injective maps into injective maps and has a right adjoint, namely

$$u : M \mapsto u(M) = \{x \in M \mid \text{stabilizer of } x \text{ is open}\}$$

Since $v(M) = 0 \Rightarrow M = 0$ we conclude by Homology, Lemma 0161. □